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Breathable cushion

Abstract

A breathable cushion in accordance with the present disclosure includes a frame arranged inside a cover. The frame includes an outer rim, an inner rim spaced apart from the outer rim, and a lattice extending between the outer rim and the inner rim and defining an internal void of the frame. Apertures are formed through the lattice of the frame. The apertures and internal void allow airflow through the frame.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to cushions, and more specifically to breathable cushions for supporting a user's head or softening impacts on a supporting surface.

BACKGROUND

(2) Cushions containing filling material reduce or eliminate airflow through the cushion. The filling material can also trap unwanted contaminants, and make washing or maintenance of the cushion difficult. Accordingly, there is a need for a cushion having breathability and ease of maintenance.

SUMMARY

(3) According

rim, and a lattice extending between the outer rim and the inner rim. The lattice may include a first webbing extending between the outer rim and the inner rim and a second webbing extending between the outer rim and the inner rim.

(4) In illustrative embodiments, the second webbing may be spaced apart from the first webbing to define an internal void of the breathable cushion frame.

(5) In illustrative embodiments, the first and second webbings may each include a plurality of interconnected web members defining a plurality of apertures through the lattice.

(6) In illustrative embodiments, the internal void may be continuous around the inner rim, and the inner rim may define an opening through the breathable cushion frame.

(7) In illustrative embodiments, the frame may be arranged inside a cover.

(8) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings disclose exemplary embodiments in which like reference characters designate the same or similar parts throughout the figures, of which:

(2) FIG. 1 is a perspective view of a bed having a support structure with walls and a mattress arranged on the support structure and showing various embodiments of breathable cushions in accordance with the present disclosure arranged on the bed;

(3) FIG. 2 is a perspective view of one embodiment of a breathable cushion frame in accordance with the present disclosure showing that the frame includes inner and outer rims and a lattice extending between the inner and outer rims;

(4) FIG. 3 is sectional view taken along line 3-3 in FIG. 2 showing that the lattice includes a first webbing extending between the inner and outer rims and a second webbing extending between the inner and outer rims and spaced apart from the first webbing to define an internal void of the breathable cushion frame around the inner rim, and that each of the first and second webbings includes a plurality of interconnected web members defining a plurality of apertures through the lattice;

(5) FIG. 4 is a top plan view of the breathable cushion frame of FIG. 1 showing that the apertures are arranged in offset repeating patterns around the inner rim and that apertures arranged along patterns closer to the inner rim have smaller opening areas than apertures arranged farther from the inner rim;

(6) FIG. 5 is sectional view taken along line 5-5 in FIG. 4 showing that in the illustrative embodiment the inner rim is arranged at a first distance away from the outer rim toward a rear of the breathable cushion frame and at a second shorter distance from the outer rim toward a front of the frame

- and a cross sectional area of a rear section of the frame is larger than a cross sectional area of a front section of the frame along a centerline of the frame;
- (7) FIG. 6 is a right-side elevation view of the breathable cushion frame of FIG. 1 showing that the breathable cushion frame tapers from front to rear at a taper angle;
- (8) FIG. 7 is a front elevation view of the breathable cushion frame of FIG. 1 showing that the front section reduces in cross sectional area from left and right toward the centerline of the frame;
- (9) FIG. 8 is an exploded perspective assembly view of the breathable cushion frame of FIG. 1 showing that in the illustrative embodiment the frame includes a first shell including the first webbing and portions of the inner and outer rims and a second shell including the second webbing and portions of the inner and outer rims, and suggesting that the first shell is coupled to the second shell along the portions of the inner and outer rims;
- (10) FIG. 9 is an enlarged view of the breathable cushion frame of FIG. 8 showing that complementary detents are formed on the outer rim portions for alignment of the outer rim portions of the first and second shells;
- (11) FIG. 10 is an enlarged view of the breathable cushion frame of FIG. 8 showing that complementary detents are formed on the inner rim portions for alignment of the inner rim portions of the first and second shells;
- (12) FIG. 11 is a perspective view of another embodiment of a breathable cushion frame in accordance with the present disclosure showing that the frame includes inner and outer rims and a lattice extending between the inner and outer rims;
- (13) FIG. 12 is sectional view taken along line 12-12 in FIG. 11 showing that the lattice includes a first webbing extending between the inner and outer rims and a second webbing extending between the inner and outer rims and spaced apart from the first webbing to define an internal void of the breathable cushion frame around the inner rim, and that each of the first and second webbings includes a plurality of interconnected web members defining a plurality of apertures through the lattice;
- (14) FIG. 13 is a top plan view of the breathable cushion frame of FIG. 11 showing that the apertures are arranged in offset repeating patterns around the inner rim and that apertures arranged along patterns closer to the inner rim have smaller opening areas than apertures arranged farther from the inner rim;
- (15) FIG. 14 is sectional view taken along line 14-14 in FIG. 13 and showing that the first webbing extends substantially along a first plane between the inner and outer rims and the second webbing extends between the inner and outer rims substantially along a second plane spaced apart from the first plane;
- (16) FIG. 15 is a right-side elevation view of the breathable cushion frame of FIG. 11;
- (17) FIG. 16 is a front elevation view of the breathable cushion frame of FIG. 11;
- (18) FIG. 17 is an exploded perspective assembly view of the breathable cushion frame of FIG. 11 showing that in the illustrative embodiment the frame includes a first shell including the first webbing and portions of the inner and outer rims and a second shell including the second webbing and portions of the inner and outer rims, and suggesting that the first shell is coupled to the second shell along the portions of the inner and outer rims;
- (19) FIG. 18 is an enlarged view of the breathable cushion frame of FIG. 17 showing that complementary detents are formed on the outer rim portions for alignment of the outer rim portions of the first and second shells;
- (20) FIG. 19 is an enlarged view of the breathable cushion frame of FIG. 17 showing that complementary detents are formed on the inner rim portions for alignment of the inner rim portions of the first and second shells;
- (21) FIG. 20 is an exploded perspective assembly view of another embodiment of a breathable cushion frame in accordance with the present disclosure showing that the frame includes a first shell and a second shell coupled together by a plurality of connectors arranged along inner and outer rims of the frame; and
- (22) FIG. 21 is an enlarged view of the breathable cushion frame of FIG. 20 showing that each connector includes a receptacle and a fastener that extends into and engages with the receptacle to hold the first shell to the second shell.

DETAILED DESCRIPTION

- (23) As used in the following description, the terms “horizontal”, “vertical”, “left”, “right”, “up” and “down”, “upper” and “lower”, “inner” and “outer”, “front” and “rear”, “top” and “bottom” as well as adjectival and adverbial derivatives thereof (for example, “horizontally”, “upwardly”, or the like), simply refer to the orientation of the illustrated structure and are not intended to be limiting to the scope of the disclosure.
- (24) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.
- (25) Illustrative breathable cushions 10, 110 in accordance with the present disclosure are shown in FIG. 1. The breathable cushion 10 includes a frame 12 arranged inside a cover 14. In the illustrative embodiment, the cover 14 is a flexible, breathable material. In some embodiments, the cover 14 is formed from woven or non-woven fabric, such as muslin cotton or mesh fabric. In some embodiments, a passage is formed through the cover 14 to allow insertion and removal of the frame 12 from the cover 14, and the passage can be closed with snap fasteners or a zipper, for example.
- (26) In the exemplary embodiment, the breathable cushion 110 includes a frame 112 arranged inside a cover 114 as shown in FIG. 1. The cover 114 is a flexible, breathable material. In some embodiments, the cover 114 is formed from woven or non-woven fabric, such as muslin cotton or mesh fabric. In some embodiments, a passage is formed through the cover 114 to allow insertion and removal of the frame 112 from the cover 114, and the passage can be closed with snap fasteners or a zipper, for example. In some embodiments, the breathable cushion 110 is sized to be shorter than the walls 1006. In some embodiments, the breathable cushion 110 is sized to substantially extend from the mattress 1004 to a top rail of the wall 1006.
- (27) FIG. 1 illustratively shows the breathable cushions 10, 110 arranged on a bed 1000, such as a crib. In the exemplary embodiment, the bed 1000 includes a support structure 1002 having walls 1006 and a mattress 1004 arranged on the support structure 1002. In some embodiments, the breathable cushion 10 is configured to support the head of a user, such as an infant, laying on the mattress 1004. In some embodiments, the breathable cushion 110 is arranged along the wall 1006, or another surface, and configured to soften impacts on the wall 1006 or other supporting surface. In some embodiments, the cover 114 includes ties for securing the breathable cushion 110 on the wall 1006.
- (28) The illustrative breathable cushion frame 12 includes an outer rim 16, an inner rim 18, and a lattice 11 extending between the outer and inner rims 16, 18 as shown in FIG. 2. In some embodiments, the lattice 11 includes a first webbing 41 extending between the outer rim 16 and the inner rim 18 and a second webbing 51 extending between the outer rim 16 and the inner rim 18 and spaced apart from the first webbing 41 to define an internal void 13 of the breathable cushion frame 12 around the inner rim 18 as shown in FIGS. 2 and 3. The first webbing 41 and second webbing 51 are arranged closer together adjacent the outer and inner rims 16, 18 and farther apart between the outer and inner rims 16, 18 as shown in FIGS. 3 and 5.
- (29) Each of the first and second webbings 41, 51 includes a plurality of interconnected web members 30, 34, respectively, defining a plurality of apertures 32, 36, respectively, through the lattice 11. The apertures 32, 36 and internal void 13 allow free airflow through the frame 12 to maximize breathability of the cushion 10. In some embodiments, the internal void 13 is free of bulk filling material, such as down or feathers, natural or synthetic fiber fill, foam fill, beads, or other filling materials that can inhibit airflow through the frame 12. In some embodiments, the internal void 13 is continuous around the inner rim 18. In some embodiments, the frame 12 defines a continuous tube extending around the opening 19.
- (30) In the illustrative embodiment, the inner rim 18 defines an opening 19 through the frame 12 as shown in FIG. 2. The frame 12 defines a rear section 22, a front section 24, a left-side section 26, and a right-side section 28 around the opening 19. The front section 24 is arranged at an opposite side of the opening 19 from the rear section 22, the left-side section 26 extends between the rear and front sections 22, 24, and the right-side section 28 extends between the rear and front sections 22, 24 at an opposite side of the opening 19 from the left-side section 26. In some embodiments, the frame 12 is symmetrical left to right and/or top to bottom across a centerline C of the frame 12 extending through the rear and front sections 22, 24 as

shown in FIGS. 4 and 5. The outer and inner rims **16**, **18** are arranged along a common plane extending along the centerline C. In some embodiments, the outer rim **16** is arranged along a first plane and the inner rim **18** is arranged along a second plane spaced apart from the first plane. In some embodiments, the outer rim **16** is oblong along a first direction (e.g., perpendicular to the centerline C), and the inner rim **18** is oblong along a second direction substantially perpendicular to the first direction (e.g., along the centerline C) as shown in FIG. 4. The outer and inner rims **16**, **18** can assume other geometric and non-geometric shapes as contemplated by the present disclosure.

(31) The apertures **32**, **36** are arranged along the lattice **11** as shown in FIGS. 3-5 and 8. In some embodiments, the apertures **32**, **36** are distributed along the lattice **11** in offset repeating patterns around the inner rim **18**. The patterns can be geometrical or non-geometrical shapes, for example. In the illustrative embodiment, the patterns are circular, and successively reduce in diameter from the outer rim **16** to the inner rim **18**. Exemplary first and second patterns P.sub.1, P.sub.2 are shown in FIG. 4. The apertures **32**, **36** distributed along the pattern P.sub.1 are offset circumferentially from the apertures **32**, **36** distributed along the pattern P.sub.2. The apertures **32**, **36** distributed along the pattern P.sub.1 are also arranged closer to the inner rim **18** than the apertures **32**, **36** distributed along the pattern P.sub.2. In some embodiments, the apertures **32**, **36** in successive patterns are aligned. In some embodiments, the apertures **32** are arranged along different patterns than the apertures **36**. In some embodiments, the apertures **32**, **36** are arranged evenly within each pattern.

(32) In the illustrative embodiment, apertures **32**, **36** arranged closer to the inner rim **18** have smaller opening areas than apertures **32**, **36** arranged farther from the inner rim **18** as shown in FIGS. 4 and 8. In some embodiments, apertures **32**, **36** distributed along patterns arranged closer to the inner rim **18** have smaller opening areas than apertures **32**, **36** distributed along patterns arranged farther from the inner rim **18**. Exemplary opening areas A.sub.1, A.sub.2, A.sub.3 of various apertures **32**, **36** arranged successively closer to the inner rim **18** are shown in FIG. 4. The opening area A.sub.3 is smaller than the opening area A.sub.2, which is smaller than the opening area A.sub.1. In some embodiments, the apertures **32**, **36** are oblong with opposing points arranged toward the outer and inner rims **16**, **18** as shown in FIG. 4. In some embodiments, the apertures **32**, **36** have other geometric or non-geometric shapes.

(33) In some embodiments, the inner rim **18** is arranged at a first distance D.sub.1 away from the outer rim **16** in the rear section **22** of the frame **12** and at a second shorter distance D.sub.2 from the outer rim **16** in the front section **24** of the frame **12** as shown in FIG. 5. In the exemplary embodiment, a cross sectional area A.sub.4 of the rear section **22** is larger than a cross sectional area A.sub.5 of the front section **24** along the centerline C of the frame **12**. In some embodiments, the frame **12** tapers at a taper angle α as the frame **12** extends from the front section **24** toward the rear section **22** as shown in FIG. 6. In some embodiments, the taper angle α is less than 10°. In some embodiments, the front section **24** reduces in cross sectional area from left and right toward the centerline C of the frame **12** as shown in FIG. 7.

(34) In some embodiments, the frame **12** includes a first shell **40** and a second shell **50** coupled to the first shell **40** as shown in FIG. 8. In the illustrative embodiment, the first shell **40** includes the first webbing **41** and portions **46**, **48** of the outer and inner rims **16**, **18**, respectively, with the inner rim portion **48** defining an opening **49** through the first shell **40**. The second shell **50** includes the second webbing **51** and portions **56**, **58** of the outer and inner rims **16**, **18**, respectively, with the inner rim portion **58** defining an opening **59** through the second shell **50**. In some embodiments, the shells **40**, **50** are substantially identical and symmetrical to one another, and arranged to position the outer and inner rim portions **146**, **148** toward the outer and inner rim portions **156**, **158**.

(35) In some embodiments, complementary detents **42**, **52** and **44**, **54** are formed on the outer rim portions **46**, **56**, respectively, for alignment of the outer rim portions **46**, **56** of the first and second shells **40**, **50** as shown in FIGS. 8 and 9. For example, the detent **42** may be a recess formed into the outer rim portion **46** and the detent **52** may be a protrusion extending from the outer rim portion **56** and configured to be received in the recess of the detent **42** with the first shell **40** coupled to the second shell **50**. The detents **52**, **54** may be the same or different, with the detents **42**, **44** being the opposite complement to the detents **52**, **54**, respectively. In some embodiments, complementary detents **45**, **55** and **47**, **57** are formed on the inner rim portions **48**, **58**, respectively, for alignment of the inner rim portions **48**, **58** of the first and second shells **40**, **50** as shown in FIGS. 8 and 10. For example, the detent **45** may be a recess formed into the inner rim portion **48** and the detent **55** may be a protrusion extending from the inner rim portion **58** and configured to be received in the recess of the detent **45** with the first shell **40** coupled to the second shell **50**. The detents **55**, **57** may be the same or different, with the detents **45**, **47** being the opposite complement to the detents **55**, **57**, respectively.

(36) In some embodiments, the first shell **40** is formed as a unitary and integral structure and the second shell **50** is formed as a separate unitary and integral structure. In some embodiments, the first shell **40** is coupled to the second shell **50** along the outer rim portions **46**, **56** and the inner rim portions **48**, **58** by an adhesive, ultrasonic welding, or heat press process among other processes contemplated by the present disclosure. In some embodiments, the frame **12** is formed as a unitary and integral structure. In some embodiments, the outer rim **16** and/or inner rim **18** are segmented and/or removed entirely such that the first and second webbings **41**, **51** engage with one another along the various sections **22**, **24**, **26**, **28** of the frame **12** to define the lattice **11** and internal void **113**.

(37) In some embodiments, the frame **12** is formed of a flexible, resilient material. In some embodiments, the frame **12** is formed from a food-grade thermoplastic polyurethane (TPU) material. In some embodiments, the material used to form the frame **12** has a Shore A durometer hardness of about 75 to about 90. In some embodiments, the frame **12** is molded, such as using an injection, cast, or compression molding process.

(38) According to other embodiments, the illustrative breathable cushion frame **112** includes an outer rim **116**, an inner rim **118**, and a lattice **111** extending between the outer and inner rims **116**, **118** as shown in FIG. 11. In some embodiments, the lattice **111** includes a first webbing **141** extending between the outer rim **116** and the inner rim **118** and a second webbing **151** extending between the outer rim **116** and the inner rim **118** and spaced apart from the first webbing **141** to define an internal void **113** of the breathable cushion frame **112** around the inner rim **118** as shown in FIGS. 11 and 12. The first webbing **141** and second webbing **151** are arranged closer together adjacent the outer and inner rims **116**, **118** and farther apart between the outer and inner rims **116**, **118** as shown in FIGS. 12 and 14.

(39) Each of the first and second webbings **141**, **151** includes a plurality of interconnected web members **130**, **134**, respectively, defining a plurality of apertures **132**, **136**, respectively, through the lattice **111**. The apertures **132**, **136** and internal void **113** allow free airflow through the frame **112** to maximize breathability of the cushion **110**. In some embodiments, the internal void **113** is free of bulk filling material, such as down or feathers, natural or synthetic fiber fill, foam fill, beads, or other filling materials that can inhibit airflow through the frame **112**. In some embodiments, the internal void **113** is continuous around the inner rim **118**. In some embodiments, the frame **112** defines a continuous tube extending around the opening **119**.

(40) In the illustrative embodiment, the inner rim **118** defines an opening **119** through the frame **112** as shown in FIG. 11. The frame **112** defines a rear section **122**, a front section **124**, a left-side section **126**, and a right-side section **128** around the opening **119**. The front section **124** is arranged at an opposite side of the opening **119** from the rear section **122**, the left-side section **126** extends between the rear and front sections **122**, **124**, and the right-side section **128** extends between the rear and front sections **122**, **124** at an opposite side of the opening **119** from the left-side section **126**. In some embodiments, the frame **112** is symmetrical left to right and/or top to bottom across a centerline C of the frame **112** extending through the rear and front sections **122**, **124** as shown in FIGS. 13 and 14. In some embodiments, the first webbing **141** extends substantially along a first plane between the outer and inner rims **116**, **118**, and the second webbing **151** extends substantially along a spaced apart second plane between the outer and inner rims **116**, **118** as shown in FIGS. 11 and 14-17. In some embodiments, the outer and inner rims **116**, **118** are arranged along a common third plane extending along the centerline C. In some embodiments, the outer rim **116** is arranged along a fourth plane and the inner rim **118** is arranged along a fifth plane spaced apart from the fourth plane. In some embodiments, the outer rim **116** is substantially rectangular as shown in FIG. 11. The outer and inner rims **116**, **118** can assume other geometric, such as square, and non-geometric shapes as contemplated by the present disclosure.

(41) The apertures **132**, **136** are arranged along the lattice **111** as shown in FIGS. 12-14 and 17. In some embodiments, the apertures **132**, **136** are distributed along the lattice **111** in offset repeating patterns around the inner rim **118**. The patterns can be geometrical or non-geometrical shapes, for

example. In the illustrative embodiment, the patterns are circular and successively reduce in diameter from the outer rim **116** to the inner rim **118**. Exemplary first and second patterns P.sub.1, P.sub.2 are shown in FIG. 13. The apertures **132**, **136** distributed along the pattern P.sub.1 are offset circumferentially from the apertures **132**, **136** distributed along the pattern P.sub.2. The apertures **132**, **136** distributed along the pattern P.sub.1 are also arranged closer to the inner rim **118** than the apertures **132**, **136** distributed along the pattern P.sub.2. In some embodiments, the apertures **132**, **136** in successive patterns are aligned. In some embodiments, the apertures **132** are arranged along different patterns than the apertures **136**. In some embodiments, the apertures **132**, **136** are arranged evenly within each pattern.

(42) In the illustrative embodiment, apertures **132**, **136** arranged closer to the inner rim **118** have smaller opening areas than apertures **132**, **136** arranged farther from the inner rim **118** as shown in FIGS. 13 and 17. In some embodiments, apertures **132**, **136** distributed along patterns arranged closer to the inner rim **118** have smaller opening areas than apertures **132**, **136** distributed along patterns arranged farther from the inner rim **118**. Exemplary opening areas A.sub.1, A.sub.2, A.sub.3 of various apertures **132**, **136** arranged successively closer to the inner rim **118** are shown in FIG. 13. The opening area A.sub.3 is smaller than the opening area A.sub.2, which is smaller than the opening area A.sub.1. In some embodiments, the apertures **132**, **136** are oblong with opposing points arranged toward the outer and inner rims **116**, **118** as shown in FIG. 13. In some embodiments, the apertures **132**, **136** have other geometric or non-geometric shapes.

(43) In some embodiments, the frame **112** includes a first shell **140** and a second shell **150** coupled to the first shell **140** as shown in FIG. 17. In the illustrative embodiment, the first shell **140** includes the first webbing **141** and portions **146**, **148** of the outer and inner rims **116**, **118**, respectively, with the inner rim portion **148** defining an opening **149** through the first shell **140**. The second shell **150** includes the second webbing **151** and portions **156**, **158** of the outer and inner rims **116**, **118**, respectively, with the inner rim portion **158** defining an opening **159** through the second shell **150**. In some embodiments, the shells **140**, **150** are substantially identical and symmetrical to one another, and arranged to position the outer and inner rim portions **146**, **148** toward the outer and inner rim portions **156**, **158**.

(44) In some embodiments, complementary detents **142**, **152** and **144**, **154** are formed on the outer rim portions **146**, **156**, respectively, for alignment of the outer rim portions **146**, **156** of the first and second shells **140**, **150** as shown in FIGS. 17 and 18. For example, the detent **142** may be a recess formed into the outer rim portion **146** and the detent **152** may be a protrusion extending from the outer rim portion **156** and configured to be received in the recess of the detent **142** with the first shell **140** coupled to the second shell **150**. The detents **152**, **154** may be the same or different, with the detents **142**, **144** being the opposite complement to the detents **152**, **154**, respectively. In some embodiments, complementary detents **145**, **155** and **147**, **157** are formed on the inner rim portions **148**, **158**, respectively, for alignment of the inner rim portions **148**, **158** of the first and second shells **140**, **150** as shown in FIGS. 17 and 19. For example, the detent **145** may be a recess formed into the inner rim portion **148** and the detent **155** may be a protrusion extending from the inner rim portion **158** and configured to be received in the recess of the detent **145** with the first shell **140** coupled to the second shell **150**. The detents **155**, **157** may be the same or different, with the detents **145**, **147** being the opposite complement to the detents **155**, **157**, respectively. In the illustrative embodiment, the inner rim **118** defines a column (formed from inner rim portions **148**, **158**) extending between the first and second webbings **141**, **151** as shown in FIGS. 12, 14, 17, and 19.

(45) In some embodiments, the first shell **140** is formed as a unitary and integral structure and the second shell **150** is formed as a separate unitary and integral structure. In some embodiments, the first shell **140** is coupled to the second shell **150** along the outer rim portions **146**, **156** and the inner rim portions **148**, **158** by an adhesive, ultrasonic welding, or heat press process among other processes contemplated by the present disclosure. In some embodiments, the frame **112** is formed as a unitary and integral structure. In some embodiments, the outer rim **116** and/or inner rim **118** are segmented and/or removed entirely such that the first and second webbings **141**, **151** engage with one another along the various sections **122**, **124**, **126**, **128** of the frame **112** to define the lattice **111** and internal void **113**.

(46) In some embodiments, the frame **112** is formed of a flexible, resilient material. In some embodiments, the frame **112** is formed from a food-grade thermoplastic polyurethane (TPU) material. In some embodiments, the material used to form the frame **112** has a Shore A durometer hardness of about 75 to about 90. In some embodiments, the frame **112** is molded, such as using an injection, cast, or compression molding process.

(47) Another exemplary embodiment of a breathable cushion frame **212** in accordance with the present disclosure is shown in FIGS. 20 and 21. The frame **212** is similar to the frame **12** shown in FIGS. 1-10, and similar numbers in the 200's are used to identify similar portions. At least one difference between the frame **212** and the frame **12** is that the first and second shells **240**, **250** of the frame **212** are coupled together by connectors arranged around the outer and inner rims **216**, **218**. In the illustrative embodiment, each connector includes a receptacle (e.g., **262**, **266**, **272**, **276**) coupled to one of the first or second shells **240**, **250** and a fastener (e.g., **264**, **268**, **274**, **278**) coupled to the other of the first or second shells **240**, **250**. Each receptacle includes a base **271** and a hole **273** formed through the base **271** as shown in FIG. 21. Each fastener includes a base **275**, a stem **277** extending from the base **275**, and a head **279** coupled to the stem **277** opposite of the base **275**. The fasteners extend into and engage with the receptacles to hold the first shell **240** to the second shell **250**. In some embodiments, the hole **273** of the receptacle has a first diameter, the stem **277** of the fastener has a second diameter substantially the same or smaller than the first diameter, the head **279** of the fastener has a third diameter larger than the first diameter. The head **279** extends through the hole **273** and engages with the base **271** of the receptacle to hold the stem **277** of the fastener in the hole **273** of the receptacle. In some embodiments, a mix of receptacles and fasteners are arranged on the first shell **240** with a complementary mix of receptacles and fasteners arranged on the second shell **250**. In some embodiments, all fasteners are arranged on one of the first or second shells **240**, **250**, and all of the receptacles are arranged on the other of the first or second shells **240**, **250**. In some embodiments, similar connectors are used to couple together the shells **140**, **150** of the frame **112** shown in FIGS. 11-19.

(48) Some non-limiting examples of the above-described embodiments can include the following:

(49) 1. A breathable cushion frame, comprising: an outer rim; an inner rim spaced apart from the outer rim; and a lattice having a first webbing extending between the outer rim and the inner rim and a second webbing extending between the outer rim and the inner rim, wherein the second webbing is spaced apart from the first webbing to define an internal void of the breathable cushion frame, and wherein each of the first and second webbings includes a plurality of interconnected web members defining a plurality of apertures through the lattice.

(50) 2. The breathable cushion frame of 1, wherein the inner rim and outer rim are arranged along a common plane.

(51) 3. The breathable cushion frame of 1, wherein the internal void is continuous around the inner rim.

(52) 4. The breathable cushion frame of 1, wherein inner rim defines an opening through the breathable cushion frame.

(53) 5. The breathable cushion frame of 4, wherein the frame defines a continuous tube extending around the opening.

(54) 6. The breathable cushion frame of 4, wherein the frame defines a rear section, a front section arranged at an opposite side of the opening from the rear section, a right-side section extending between the front and rear sections, and a left-side section extending between the front and rear sections at an opposite side of the opening from the right-side section.

(55) 7. The breathable cushion frame of 6, wherein the left-side section and right-side section are substantially symmetrical across a centerline extending through the front and rear sections.

(56) 8. The breathable cushion frame of 7, wherein the rear section has a larger cross sectional area than the front section along a centerline extending through the front and rear sections.

(57) 9. The breathable cushion frame of 7, wherein the inner rim is arranged at a first distance from the outer rim to define the rear section and a second distance from the outer rim to define the front section.

(58) 10. The breathable cushion frame of 9, wherein second distance is smaller than the first distance.

(59) 11. The breathable cushion frame of 7, wherein the frame is tapered from the front section to the rear section at a taper angle.

(60) 12. The breathable cushion frame of 7, wherein the taper angle less than 10 degrees.

(61) 13. The breathable cushion frame of 7, wherein the front section reduces in cross sectional area from left and right toward a centerline extending

through the front and rear sections.

(62) 14. The breathable cushion frame of 1, wherein the first webbing and second webbing are arranged closer together adjacent the inner and outer rims and farther apart between the inner and outer rims.

(63) 15. The breathable cushion frame of 1, wherein the first webbing extends substantially along a first plane between the inner and outer rims, wherein the second webbing extends substantially along a second plane between the inner and outer rims, and wherein the first plane is spaced apart from the second plane.

(64) 16. The breathable cushion frame of 11, wherein the apertures are distributed along the lattice in offset repeating patterns around the inner rim.

(65) 17. The breathable cushion frame of 16, wherein apertures arranged along patterns closer to the inner rim have smaller opening areas than apertures arranged farther from the inner rim.

(66) 18. The breathable cushion frame of 17, wherein the patterns are circular and successively smaller in diameter moving from the outer rim to the inner rim.

(67) 19. The breathable cushion frame of 16, wherein, within each pattern, the apertures are arranged evenly.

(68) 20. The breathable cushion frame of 1, wherein the apertures are oblong with opposing points arranged toward the inner and outer rims.

(69) 21. The breathable cushion frame of 1, wherein the outer rim is substantially rectangular.

(70) 22. The breathable cushion frame of 21, wherein the outer rim is substantially square.

(71) 23 The breathable cushion frame of 1, wherein the outer rim is oblong along a first direction, and wherein the inner rim is oblong along a second direction substantially perpendicular to the first direction.

(72) 24. The breathable cushion frame of 1, further comprising a first shell defining the first webbing and first portions of the inner and outer rims and a second shell defining the second webbing and second portions of the inner and outer rims.

(73) 25. The breathable cushion frame of 24, wherein the first and second portions of the inner rim are coupled by ultrasonic welding, and wherein the first and second portions of the outer rim are coupled by ultrasonic welding.

(74) 26. The breathable cushion frame of 24, further comprising a plurality of connectors arranged along the inner and outer rims, wherein the connectors are configured to couple the first shell to the second shell.

(75) 27. The breathable cushion frame of 26, wherein the connectors each include a receptacle coupled to one of the first or second shells and a fastener coupled to the other of the first or second shells, and wherein the fastener extends into and engages with the receptacle to hold the first shell to the second shell.

(76) 28. The breathable cushion frame of 29, wherein each receptacle defines a hole having a first diameter, wherein each fastener includes a stem and a head coupled to the stem, wherein the stem has a second diameter substantially the same or smaller than the first diameter, and wherein the head has a third diameter larger than the first diameter.

(77) 29. The breathable cushion frame of 24, further comprising complimentary detents on the inner and outer rims for alignment of the first and second shells.

(78) 30. The breathable cushion frame of 29, wherein the first and second shells are symmetrical to one another.

(79) 31. The breathable cushion frame of 24, wherein the first webbing and first portions of the inner and outer rims are unitary and integral, and wherein the second webbing and second portions of the inner and outer rims are unitary and integral.

(80) 32. The breathable cushion frame of 1, wherein the lattice, inner rim, and outer rim are unitary and integral.

(81) 33. The breathable cushion frame of 1, wherein the frame is formed from a flexible, resilient material.

(82) 34 The breathable cushion frame of 33, wherein the frame is formed from thermoplastic polyurethane.

(83) 35. The breathable cushion frame of 33, wherein the material has a Shore A durometer hardness of about 75 to about 90.

(84) 36. The breathable cushion frame of 1, wherein the inner rim defines a column.

(85) 37. A breathable cushion, comprising: a cover; and the breathable cushion frame of 1 arranged inside the cover.

(86) 38. The breathable cushion of 37, wherein the cover is a fabric material.

(87) 39. A breathable cushion frame, comprising: a lattice having a first webbing and a second webbing, each of the first and second webbings includes a plurality of interconnected web members defining a plurality of apertures through the lattice, wherein the second webbing is spaced apart from the first webbing to define an internal void of the breathable cushion frame, and wherein the lattice extends around an opening through the breathable cushion frame to define a continuous tube extending around the opening.

(88) The variations and benefits of each embodiment described herein are equally applicable to the other embodiments described herein, and any combinations therebetween are fully contemplated by the present disclosure.

(89) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A breathable cushion frame, comprising: an outer rim; an inner rim spaced apart from the outer rim; and a lattice having a first webbing extending between the outer rim and the inner rim and a second webbing extending between the outer rim and the inner rim, wherein a first shell defines the first webbing and first portions of the inner and outer rims and a second shell defines the second webbing and second portions of the inner and outer rims, wherein a plurality of connectors are arranged along the inner and outer rims and configured to couple the first shell to the second shell, wherein the connectors each include a receptacle coupled to one of the first or second shells and a fastener coupled to the other of the first or second shells, wherein the fastener extends into and engages with the receptacle to hold the first shell to the second shell, wherein the second webbing is spaced apart from the first webbing to define an internal void of the breathable cushion frame, and wherein each of the first and second webbings includes a plurality of interconnected web members defining a plurality of apertures through the lattice.

2. The breathable cushion frame of claim 1, wherein the internal void is continuous around the inner rim.

3. The breathable cushion frame of claim 1, wherein inner rim defines an opening through the breathable cushion frame.

4. The breathable cushion frame of claim 3, wherein the breathable cushion frame defines a continuous tube extending around the opening.

5. The breathable cushion frame of claim 1, wherein the first webbing and second webbing are arranged closer together adjacent the inner and outer rims and farther apart between the inner and outer rims.

6. The breathable cushion frame of claim 1, wherein the first webbing extends substantially along a first plane between the inner and outer rims, wherein the second webbing extends substantially along a second plane between the inner and outer rims, and wherein the first plane is spaced apart from the second plane.

7. The breathable cushion frame of claim 1, wherein the plurality of apertures are distributed along the lattice in offset repeating patterns around the inner rim.

8. The breathable cushion frame of claim 7, wherein apertures arranged along patterns closer to the inner rim have smaller opening areas than apertures arranged farther from the inner rim.

9. The breathable cushion frame of claim 8, wherein the patterns are circular and successively smaller in diameter moving from the outer rim to the inner rim.

10. The breathable cushion frame of claim 1, wherein the outer rim is substantially rectangular.
 11. The breathable cushion frame of claim 1, wherein the outer rim is oblong along a first direction, and wherein the inner rim is oblong along a second direction substantially perpendicular to the first direction.
 12. A breathable cushion, comprising: a cover; and the breathable cushion frame of claim 1 arranged inside the cover.
 13. The breathable cushion of claim 12, wherein the cover is a fabric material.
 14. The breathable cushion frame of claim 1, wherein each receptacle defines a hole having a first diameter, wherein each fastener includes a stem and a head coupled to the stem, wherein the stem has a second diameter the same or smaller than the first diameter, and wherein the head has a third diameter larger than the first diameter.
 15. The breathable cushion frame of claim 1, wherein the first and second shells are symmetrical to one another.
 16. The breathable cushion frame of claim 1, wherein the first webbing and first portions of the inner and outer rims are unitary and integral, and wherein the second webbing and second portions of the inner and outer rims are unitary and integral.
 17. The breathable cushion frame of claim 1, wherein the first and second inner rim portions together define the inner rim, wherein inner rim defines an opening through the breathable cushion frame, and wherein the internal void is continuous around the inner rim.
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